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A two-grid method for coupled free flow with porous media flow

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ABSTRACT

This paper presents a two-grid method for solving systems of partial differential equations modelling incompressible free flow coupled with porous media flow. This work considers both the coupled Stokes and Darcy as well as the coupled Navier–Stokes and Darcy problems. The numerical schemes proposed are based on combinations of the continuous finite element method and the discontinuous Galerkin method. Numerical errors and convergence rates for solutions obtained from the two-grid method are presented. CPU times for the two-grid algorithm are shown to be significantly less than those obtained by solving the fully coupled problem.

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1. Introduction

The coupling of free flow with porous media flow has been studied extensively in the literature. Interest in this coupling is mostly due to the wide range of applications for example in biosciences, geosciences and industrial applications. The incompressible Navier–Stokes and Darcy coupling has been analyzed for example in [1–4]. The Stokes and Darcy model has been studied for example in [5–14]. The coupled Navier–Stokes and Darcy schemes studied in [1,4] result in large non-linear coupled systems which may be time consuming to solve.

In this work, a decoupling strategy based on the two-grid approach is used for both Stokes/Darcy and Navier–Stokes/Darcy problems. The two-grid technique has been widely used in the literature for nonlinear problems (see for instance [15–21]). We show that the two-grid method is attractive for these systems because it decouples the coupled model into two relatively smaller problems. The first is an Oseen problem (or Stokes problem) to solve for the fluid velocity and pressure in the free flow domain. The second is a modified Darcy problem to solve for the Darcy pressure in the porous medium. The two-grid technique has several advantages. First it allows for easy parallelization. Second, legacy codes exist that have been optimized to solve the Stokes, Navier–Stokes or Darcy problems separately. These codes can be used independently with the two-grid approach. In this work, we use a multi-numerics scheme that combines the continuous finite element method with the discontinuous Galerkin (DG) method. The use of the latter in the subsurface offers many advantages: local mass conservation, local and high order accuracy, robustness of the method with respect to the heterogeneities of the medium. Recently, the two-grid method applied to a first order continuous finite element method was analyzed for solving the Stokes/Darcy [22] and Navier–Stokes/Darcy problems [23].

An outline of the paper is as follows. The mathematical model and its fully coupled numerical discretization are described in Sections 2 and 3 respectively. The two-grid method is introduced in Section 4. It yields decoupled, linearized and smaller problems to solve. Numerical examples are shown in Section 5.1. Conclusions follow.

2. Fluid flow models

Let $\Omega \subset \mathbb{R}^2$ be a bounded polygonal domain partitioned into two non-overlapping subdomains Ω_1 and Ω_2. In the free flow domain Ω_1, the flow is modelled by the Stokes or Navier–Stokes equations. The domain Ω_2 denotes the porous medium region where the flow is described by Darcy's law. The free flow and porous media domains are separated by a polygonal interface Γ_{12}. First we state the equations that govern the flow in each domain. The flow in Ω_1 can be described by either the Stokes equations:

$$-\nabla \cdot (2\nu \boldsymbol{D}(\boldsymbol{u}_1)) + \nabla p_1 = \boldsymbol{f}_1, \quad \text{in } \Omega_1, \tag{1}$$
$$\nabla \cdot \boldsymbol{u}_1 = 0, \quad \text{in } \Omega_1, \tag{2}$$
$$\boldsymbol{u}_1 = 0, \quad \text{on } \Gamma_1 = \partial\Omega \cap \partial\Omega_1 \tag{3}$$

or the Navier–Stokes equations:

$$-\nabla \cdot (2\nu \boldsymbol{D}(\boldsymbol{u}_1)) + \nabla p_1 + \boldsymbol{u}_1 \cdot \nabla \boldsymbol{u}_1 = \boldsymbol{f}_1, \quad \text{in } \Omega_1, \tag{4}$$
$$\nabla \cdot \boldsymbol{u}_1 = 0, \quad \text{in } \Omega_1, \tag{5}$$
$$\boldsymbol{u}_1 = 0, \quad \text{on } \Gamma_1. \tag{6}$$

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The fluid velocity and pressure in Ω_1 are denoted by $\boldsymbol{u}_1$ and p_1 respectively. The coefficient $\nu > 0$ is the fluid kinematic viscosity, the function $\boldsymbol{f}_1$ is an external force acting on the fluid, and $\boldsymbol{D}(\boldsymbol{u}_1)$ is the rate of strain matrix:

$$\boldsymbol{D}(\boldsymbol{u}_1) = \frac{1}{2}(\nabla \boldsymbol{u}_1 + \nabla \boldsymbol{u}_1^T). \tag{7}$$

The flow in Ω_2 is modeled by Darcy's law:

$$\boldsymbol{u}_2 = -\boldsymbol{K}\nabla p_2, \quad \text{in } \Omega_2, \tag{8}$$

$$\nabla \cdot \boldsymbol{u}_2 = f_2, \quad \text{in } \Omega_2, \tag{9}$$

$$p_2 = g_{\mathrm{D}}, \quad \text{on } \Gamma_{2\mathrm{D}}, \tag{10}$$

$$\boldsymbol{K}\nabla p_2 \cdot \boldsymbol{n}_2 = g_{\mathrm{N}}, \quad \text{on } \Gamma_{2\mathrm{N}}. \tag{11}$$

We note that the boundary of the porous medium $\Gamma_2 = \partial\Omega \cap \partial\Omega_2$ is partitioned into disjoint sets $\Gamma_{2\mathrm{N}}$ and $\Gamma_{2\mathrm{D}}$, the Neumann and Dirichlet boundary respectively. We assume that $|\Gamma_{2\mathrm{D}}| > 0$. The fluid velocity and pressure in Ω_2 are denoted by $\boldsymbol{u}_2$ and p_2 respectively. The function f_2 is a source term, the functions g_{D} and g_{N} are the prescribed value and flux respectively. The vector $\boldsymbol{n}_2$ denotes the unit vector normal to Γ_2 and the coefficient $\boldsymbol{K}$ is a symmetric positive definite tensor representing the hydraulic conductivity of the fluid.

The coupled models are completed by specifying the following coupling conditions on the interface Γ_{12}:

(1) The continuity of the normal component of velocity arises from the incompressibility condition. Let $\boldsymbol{n}_{12}$ be the unit normal vector to Γ_{12} directed from Ω_1 to Ω_2.

$$\boldsymbol{u}_1 \cdot \boldsymbol{n}_{12} = -\boldsymbol{K}\nabla p_2 \cdot \boldsymbol{n}_{12}. \tag{12}$$

(2) The second condition states that the tangential component of the velocity in the free flow region is proportional to the shear stress. Let $\boldsymbol{\tau}_{12}$ be the unit tangent vector on the interface Γ_{12} then the Beavers–Joseph–Saffman law [24,25] will be written as:

$$\boldsymbol{u}_1 \cdot \boldsymbol{\tau}_{12} = -2\nu G \boldsymbol{D}(\boldsymbol{u}_1)\boldsymbol{n}_{12} \cdot \boldsymbol{\tau}_{12}. \tag{13}$$

The constant G depends on the permeability as $G = G_0(\boldsymbol{K}\boldsymbol{\tau}_{12} \cdot \boldsymbol{\tau}_{12})^{1/2}$. The multiplicative constant G_0 depends on the properties of the porous medium and can be obtained experimentally. Condition (13) has been mathematically justified by asymptotic analysis in [26,27].

(3) The last condition relates to the balance of forces on the interface:

$$-2\nu \boldsymbol{D}(\boldsymbol{u}_1)\boldsymbol{n}_{12} \cdot \boldsymbol{n}_{12} + p_1 = p_2. \tag{14}$$

3. Numerical schemes

Let $\mathcal{E}_h = \mathcal{E}_1^h \cup \mathcal{E}_2^h$ be a regular triangulation of the computational domain Ω with elements E [28]. The velocity and pressure solutions on Ω_1 belong to the conforming MINI finite element spaces $\boldsymbol{X}_1^h$ and M_1^h respectively. We recall that the space $\boldsymbol{X}_1^h$ consists of continuous piecewise linears plus cubic bubbles and the space M_1^h consists of continuous piecewise linears [29]. In Ω_1, this yields a first order method. The Darcy pressure belongs to the space of discontinuous piecewise polynomials of degree $k_2 \geqslant 1$:

$$M_2^h = \left\{ q_2 \in L^2(\Omega_2) : \forall E \in \mathcal{E}_2^h, q_2|_E \in \mathbb{P}_{k_2}(E) \right\}. \tag{15}$$

With this particular choice, the Dirichlet datum on $\Gamma_{2\mathrm{D}}$ is imposed weakly. The flexibility of discontinuous Galerkin methods also allows for test functions that satisfy the Dirichlet datum strongly. For instance if $g_{\mathrm{D}} = 0$, the space M_2^h can be chosen as

$$M_2^h = \left\{ q_2 \in L^2(\Omega_2) : \forall E \in \mathcal{E}_2^h,\ q_2|_E \in \mathbb{P}_{k_2}(E); \quad q_2 = 0, \quad \text{on } \Gamma_{2\mathrm{D}} \right\}. \tag{16}$$

We denote by Γ_2^h the set of interior edges of $\mathcal{E}_2^h$. In addition, a unit normal vector $\boldsymbol{n}_e$ is associated with each edge e. In the case that e lies on the boundary $\partial\Omega_2$, then $\boldsymbol{n}_e$ is taken to be the outward unit normal vector to $\partial\Omega_2$. For any two elements E_i and E_j (with $i < j$) of $\mathcal{E}_2^h$ that share a common edge e, we define the jump $[v]$ and average $\{v\}$ of a piecewise discontinuous polynomial v by

$$[v] = (v|_{E_i}) - (v|_{E_j}), \quad \text{and} \quad \{v\} = \frac{1}{2}(v|_{E_i}) + \frac{1}{2}(v|_{E_j}),$$

respectively. The choice of the coupling of the continuous finite element method (FEM) with the discontinuous Galerkin method is due to the fact that the continuous finite element method is adequate for the free flow region for problems considered in this work. On the other hand, the discontinuous Galerkin method is better suited to handle heterogeneities such as kinks and faults that arise in the porous medium. We refer the reader to [30] for a detailed discussion on this matter. In order to define the numerical scheme for the Stokes/Darcy and Navier–Stokes/Darcy couplings, we first introduce the following bilinear forms.

- For the discretization of the viscous term in the Stokes or Navier–Stokes equations:

$$\forall \boldsymbol{v},\ \boldsymbol{w} \in \boldsymbol{X}_1^h, \quad a_{\mathrm{S}}(\boldsymbol{v},\ \boldsymbol{w}) = 2\nu(\boldsymbol{D}(\boldsymbol{v}),\ \boldsymbol{D}(\boldsymbol{w}))_{\Omega_1}.$$

- For the discretization of the pressure term in the Stokes or Navier–Stokes equations:

$$\forall \boldsymbol{v} \in \boldsymbol{X}_1^h, \quad \forall q_1 \in M_1^h, \quad b_{\mathrm{S}}(\boldsymbol{v}, q_1) = -(q_1, \nabla \cdot \boldsymbol{v})_{\Omega_1}.$$

- For the discretization of the non-linear term in the Navier–Stokes equations:

$$\forall \boldsymbol{z}, \boldsymbol{v}, \boldsymbol{w} \in \boldsymbol{X}_1^h,$$
$$c_{\mathrm{NS}}(\boldsymbol{z}, \boldsymbol{v}, \boldsymbol{w}) = \frac{1}{2}(\boldsymbol{z} \cdot \nabla \boldsymbol{v}, \boldsymbol{w})_{\Omega_1} - \frac{1}{2}(\boldsymbol{z} \cdot \nabla \boldsymbol{w}, \boldsymbol{v})_{\Omega_1} + \frac{1}{2}(\boldsymbol{z} \cdot \boldsymbol{n}_{12}, \boldsymbol{v} \cdot \boldsymbol{w})_{\Gamma_{12}}.$$

- For the discretization of the diffusion term in Darcy's law:

$$\begin{aligned} \forall q_2, t_2 \in M_2^h, \quad a_{\mathrm{D}}(q_2, t_2) = & \sum_{E \in \mathcal{E}_2^h} (\boldsymbol{K}\nabla q_2, \nabla t_2)_E \\ & + \sum_{e \in \Gamma_2^h \cup \Gamma_{2\mathrm{D}}} \frac{\sigma}{|e|}([q_2], [t_2])_e \\ & - \sum_{e \in \Gamma_2^h \cup \Gamma_{2\mathrm{D}}} (\{\boldsymbol{K}\nabla q_2 \cdot \boldsymbol{n}_e\}[t_2])_e \\ & + \epsilon_2 \sum_{e \in \Gamma_2^h \cup \Gamma_{2\mathrm{D}}} (\{\boldsymbol{K}\nabla t_2 \cdot \boldsymbol{n}_e\}, [q_2])_e. \end{aligned}$$

The parameters σ and ϵ_2 are the penalty and symmetrization parameters respectively. The case $\epsilon_2 = -1$ yields the Symmetric Interior Penalty Galerkin (SIPG) method whereas the case $\epsilon_2 = 1$ yields the Non-Symmetric Interior Penalty Galerkin (NIPG) method [31]. We also note that the form a_{D} depends strongly on the underlying mesh. We may use the notation $a_{\mathrm{D}} = a_{\mathrm{D},h}$ in the definition of the two-grid method to avoid any confusion. The natural norm associated with the form a_{D} is:

$$\forall q_2 \in M_2^h, \ \|q_2\|_{\mathrm{DG}} = \left(\sum_{E \in \mathcal{E}_2^h} \|\nabla q_2\|_{L^2(E)}^2 + \sum_{e \in \Gamma_2^h \cup \Gamma_{2\mathrm{D}}} \frac{\sigma}{|e|} \|[q_2]\|_{L^2(e)}^2 \right)^{1/2}.$$

- For the discretization of the interface terms:

$$\forall \boldsymbol{v}, \boldsymbol{w} \in \boldsymbol{X}_1^h, \quad \forall q_2, t_2 \in M_2^h,$$
$$\gamma(\boldsymbol{v}, q_2; \boldsymbol{w}, t_2) = (q_2, \boldsymbol{w}\cdot\boldsymbol{n}_{12})_{\Gamma_{12}} - (\boldsymbol{v}\cdot\boldsymbol{n}_{12}, t_2)_{\Gamma_{12}} + \frac{1}{G}(\boldsymbol{v}\cdot\boldsymbol{\tau}_{12}, \boldsymbol{w}\cdot\boldsymbol{\tau}_{12})_{\Gamma_{12}}.$$

- For the discretization of the data terms:

$$\forall t_2 \in M_2^h, \quad \forall \boldsymbol{v} \in \boldsymbol{X}_1^h,$$
$$L(\boldsymbol{v}, t_2) = (\boldsymbol{f}_1, \boldsymbol{v})_{\Omega_1} + (f_2, t_2)_{\Omega_2} + \epsilon_2 \sum_{e\in\Gamma_{2D}} (\boldsymbol{K}\nabla t_2\cdot\boldsymbol{n}_e, g_D)_e + \sum_{e\in\Gamma_{2D}} \frac{\sigma}{|e|}(g_D, t_2)_e + \sum_{e\in\Gamma_{2N}} (g_N, t_2)_e.$$

We now define the numerical schemes for the coupled Stokes/Darcy and coupled Navier–Stokes/Darcy problems $\left(W_S^h\right)$ and $\left(W_{NS}^h\right)$ respectively:

$$\left(W_S^h\right)\begin{cases} \text{Find } \boldsymbol{U}_1^h \in \boldsymbol{X}_1^h, P_1^h \in M_1^h, P_2^h \in M_2^h, \text{s.t} \\ \forall \boldsymbol{v}_1 \in \boldsymbol{X}_1^h, q_2 \in M_2^h,\ a_S\left(\boldsymbol{U}_1^h, \boldsymbol{v}_1\right) + b_S\left(\boldsymbol{v}_1, P_1^h\right) + a_D\left(P_2^h, q_2\right) \\ +\gamma\left(\boldsymbol{U}_1^h, P_2^h; \boldsymbol{v}_1, q_2\right) = L(\boldsymbol{v}_1, q_2), \\ \forall q_1 \in M_1^h,\ b_S\left(\boldsymbol{U}_1^h, q_1\right) = 0, \end{cases}$$

$$\left(W_{NS}^h\right)\begin{cases} \text{Find } \boldsymbol{U}_1^h \in \boldsymbol{X}_1^h, P_1^h \in M_1^h, P_2^h \in M_2^h, \text{s.t} \\ \forall \boldsymbol{v}_1 \in \boldsymbol{X}_1^h, q_2 \in M_2^h,\ a_S\left(\boldsymbol{U}_1^h, \boldsymbol{v}_1\right) + b_S\left(\boldsymbol{v}_1, P_1^h\right) + c_{NS}\left(\boldsymbol{U}_1^h; \boldsymbol{U}_1^h, \boldsymbol{v}_1\right) \\ +a_D\left(P_2^h, q_2\right) + \gamma\left(\boldsymbol{U}_1^h, P_2^h; \boldsymbol{v}_1, q_2\right) = L(\boldsymbol{v}_1, q_2), \\ \forall q_1 \in M_1^h,\ b_S\left(\boldsymbol{U}_1^h, q_1\right) = 0. \end{cases}$$

In order to solve the coupled problem (W_S^h), we have to assemble a large linear system of dimension $\dim \boldsymbol{X}_1^h + \dim M_1^h + \dim M_2^h$. For realistic domains this system can be very large and time consuming to solve. Similarly, solving $\left(W_{NS}^h\right)$ is time consuming as it involves a Picard iteration and large linear systems. In the next section we present a two-grid algorithm that reduces the computational cost by decoupling the coupled problems into two relatively smaller problems in the free flow and porous media domains.

4. Two grid algorithm

The objective of the two-grid method is to decouple the fully coupled problem into separate problems in the free flow and porous media domains. The technique involves two steps. First, a fully coupled problem, with relatively small cost, is solved on a coarse mesh $\mathcal{E}_H$ of size H. Second, the surface problem and subsurface problems are solved in parallel and independently on a fine mesh $\mathcal{E}_h$ of size h. For each decoupled problem, the data on the interface is obtained from the solution on the coarse mesh. The two-grid algorithm is summarized below for both linear and nonlinear cases.

4.1. Two-grid algorithm for coupled Stokes/Darcy problem

(1) First we solve a fully coupled Stokes and Darcy problem on a coarse grid of size H:

$$\left(W_S^H\right)\begin{cases} \text{Find } \boldsymbol{U}_1^H \in \boldsymbol{X}_1^H, P_1^H \in M_1^H, P_2^H \in M_2^H, \text{s.t.} \\ \forall \boldsymbol{v}_1 \in \boldsymbol{X}_1^H, \forall q_2 \in M_2^H, a_S\left(\boldsymbol{U}_1^H, \boldsymbol{v}_1\right) + b_S\left(\boldsymbol{v}_1, P_1^H\right) \\ +a_{D,H}\left(P_2^H, q_2\right) + \gamma\left(\boldsymbol{U}_1^H, P_2^H; \boldsymbol{v}_1, q_2\right) = L(\boldsymbol{v}_1, q_2), \\ \forall q_1 \in M_1^H,\ b_S\left(\boldsymbol{U}_1^H, q_1\right) = 0. \end{cases}$$

(2) We then solve a decoupled problem on a fine grid of size h:

$$(W_{h,SD})\begin{cases} \text{Find } \boldsymbol{U}_{h,1} \in \boldsymbol{X}_1^h, P_{h,1} \in M_1^h, P_{h,2} \in M_2^h, \text{s.t.} \\ \forall \boldsymbol{v}_1 \in \boldsymbol{X}_1^h, \forall q_2 \in M_2^h, a_S(\boldsymbol{U}_{h,1}, \boldsymbol{v}_1) \\ +\frac{1}{G}(\boldsymbol{U}_{h,1}\cdot\boldsymbol{\tau}_{12}, \boldsymbol{v}_1\cdot\boldsymbol{\tau}_{12})_{\Gamma_{12}} + b_S(\boldsymbol{v}_1, P_{h,1}) \\ +a_{D,h}(P_{h,2}, q_2) = L(\boldsymbol{v}_1, q_2) - (P_2^H, \boldsymbol{v}_1\cdot\boldsymbol{n}_{12})_{\Gamma_{12}} \\ +\left(\boldsymbol{U}_1^H\cdot\boldsymbol{n}_{12}, q_2\right)_{\Gamma_{12}}, \\ \forall q_1 \in M_1^h,\ b_S(\boldsymbol{U}_{h,1}, q_1) = 0. \end{cases}$$

The decoupled fine grid problem is equivalent to solving modified Stokes and Darcy problems in Ω_1 and Ω_2 respectively. The modified problems can be stated as follows: in Ω_1,

$$(W_{h,S})\begin{cases} \text{Find } \boldsymbol{U}_{h,1} \in \boldsymbol{X}_1^h, P_{h,1} \in M_1^h, \text{s.t.} \\ \forall \boldsymbol{v}_1 \in \boldsymbol{X}_1^h, \forall q_2 \in M_2^h, a_S(\boldsymbol{U}_{h,1}, \boldsymbol{v}_1) + b_S(\boldsymbol{v}_1, P_{h,1}) \\ +\frac{1}{G}(\boldsymbol{U}_{h,1}\cdot\boldsymbol{\tau}_{12}, \boldsymbol{v}_1\cdot\boldsymbol{\tau}_{12})_{\Gamma_{12}} \\ = (\boldsymbol{f}_1, \boldsymbol{v}_1)_{\Omega_1} - (P_2^H, \boldsymbol{v}_1\cdot\boldsymbol{n}_{12})_{\Gamma_{12}}, \\ \forall q_1 \in M_1^h,\ b_S(\boldsymbol{U}_{h,1}, q_1) = 0 \end{cases}$$

and in Ω_2,

$$(W_{h,D})\begin{cases} \text{Find } P_{h,2} \in M_2^h, \text{s.t.} \\ \forall q_2 \in M_2^h, a_{D,h}(P_{h,2}, q_2) = (f_2, q_2)_{\Omega_2} + (\boldsymbol{U}_1^H\cdot\boldsymbol{n}_{12}, q_2)_{\Gamma_{12}} \\ +(g_N, q_2)_{\Gamma_{2N}} \\ +\epsilon_2 \sum_{e\in\Gamma_{2D}} (\boldsymbol{K}\nabla q_2\cdot\boldsymbol{n}_e, g_D)_e + \sum_{e\in\Gamma_{2D}} \frac{\sigma}{|e|}(g_D, q_2)_e. \end{cases}$$

4.2. Two-grid algorithm for coupled Navier–Stokes and Darcy problem

(1) First we solve a nonlinear problem on a coarse grid of size H:

$$\left(W_{NS}^H\right)\begin{cases} \text{Find } \boldsymbol{U}_1^H \in \boldsymbol{X}_1^h, P_1^H \in M_1^H, P_2^H \in M_2^H, \text{s.t.} \\ \forall \boldsymbol{v}_1 \in \boldsymbol{X}_1^H, \quad \forall q_2 \in M_2^H, \quad a_S(\boldsymbol{U}_1^H, \boldsymbol{v}_1) + b_S\left(\boldsymbol{v}_1, P_1^H\right) \\ +c_{NS}\left(\boldsymbol{U}_1^H; \boldsymbol{U}_1^H; \boldsymbol{v}_1\right) \\ +a_{D,H}\left(P_2^H, q_2\right) + \gamma\left(\boldsymbol{U}_1^H, P_2^H; \boldsymbol{v}_1, q_2\right) = L(\boldsymbol{v}_1, q_2), \\ \forall q_1 \in M_1^H, \quad b_S\left(\boldsymbol{U}_1^H, q_1\right) = 0. \end{cases}$$

The solution $\left(\boldsymbol{U}_1^H, P_1^H, P_2^H\right)$ for the problem (W_{NS}^H) is obtained using a standard Picard iteration. At each iteration, given $\boldsymbol{U}_1^{H,n-1}$, we solve for $\left(\boldsymbol{U}_1^{H,n}, P_1^H, P_2^H\right)$ satisfying:

$$\begin{aligned} &\forall \boldsymbol{v}_1 \in \boldsymbol{X}_1^H, \quad \forall q_2 \in M_2^H,\ a_S\left(\boldsymbol{U}_1^{H,n}, \boldsymbol{v}_1\right) + b_S\left(\boldsymbol{v}_1, P_1^H\right) \\ &\quad + c_{NS}\left(\boldsymbol{U}_1^{H,n-1}; \boldsymbol{U}_1^{H,n}; \boldsymbol{v}_1\right) + a_{D,H}\left(P_2^H, q_2\right) \\ &\quad + \gamma(\boldsymbol{U}_1^{H,n}, P_2^H; \boldsymbol{v}_1, q_2) = L(\boldsymbol{v}_1, q_2), \\ &\quad \forall q_1 \in M_1^H,\ b_S\left(\boldsymbol{U}_1^{H,n}, q_1\right) = 0. \end{aligned}$$

We use the stopping criterion: $\left\|\boldsymbol{U}_1^{H,n}-\boldsymbol{U}_1^{H,n-1}\right\|_{L^2(\Omega_1)}\leqslant\delta$, where δ is a given tolerance.

(2) We then solve a decoupled problem on a fine grid of size h:

$$(W_{h,\mathrm{NSD}})\begin{cases}\text{Find } \boldsymbol{U}_{h,1}\in\boldsymbol{X}_1^h, P_{h,1}\in M_1^h, P_{h,2}\in M_2^h, \text{s.t.}\\ \forall\boldsymbol{v}_1\in\boldsymbol{X}_1^h,\quad\forall q_2\in M_2^h,\quad a_{\mathrm{S}}(\boldsymbol{U}_{h,1},\boldsymbol{v}_1)\\ +b_{\mathrm{S}}(\boldsymbol{v}_1,P_{h,1})+c_{\mathrm{NS}}\left(\boldsymbol{U}_1^H;\boldsymbol{U}_{h,1},\boldsymbol{v}_1\right)\\ +\frac{1}{G}(\boldsymbol{U}_{h,1}\cdot\boldsymbol{\tau}_{12},\boldsymbol{v}_1\cdot\boldsymbol{\tau}_{12})_{\Gamma_{12}}+a_{\mathrm{D},h}(P_{h,2},q_2)\\ =L(\boldsymbol{v}_1,q_2)-\left(P_2^H,\boldsymbol{v}_1\cdot\boldsymbol{n}\right)_{\Gamma_{12}}+\left(\boldsymbol{U}_1^H\cdot\boldsymbol{n}_{12},q_2\right)_{\Gamma_{12}},\\ \forall q_1\in M_1^h,\ b_{\mathrm{S}}(\boldsymbol{U}_{h,1},q_1)=0.\end{cases}$$

We note that the resulting two-grid problem $(W_{h,\mathrm{NSD}})$ does not require a Picard iteration because the non-linear term c_{NS} has been linearized. The decoupled fine grid problem is equivalent to solving modified Oseen and Darcy problems in Ω_1 and Ω_2 respectively. The modified problems can be stated as follows: in Ω_1,

$$(W_{h,\mathrm{NS}})\begin{cases}\text{Find } \boldsymbol{U}_{h,1}\in\boldsymbol{X}_1^h, P_{h,1}\in M_1^h, \text{s.t.}\\ \forall\boldsymbol{v}_1\in\boldsymbol{X}_1^h,\ \forall q_2\in M_2^h,\ a_{\mathrm{S}}(\boldsymbol{U}_{h,1},\boldsymbol{v}_1)\\ +b_{\mathrm{S}}(\boldsymbol{v}_1,P_{h,1})+c_{\mathrm{NS}}(\boldsymbol{U}_1^H;\boldsymbol{U}_{h,1},\boldsymbol{v}_1)\\ +\frac{1}{G}(\boldsymbol{U}_{h,1}\cdot\boldsymbol{\tau}_{12},\boldsymbol{v}_1\cdot\boldsymbol{\tau}_{12})_{\Gamma_{12}}=(\boldsymbol{f}_1,\boldsymbol{v}_1)_{\Omega_1}-\left(P_2^H,\boldsymbol{v}_1\cdot\boldsymbol{n}\right)_{\Gamma_{12}},\\ \forall q_1\in M_1^h,\ b_{\mathrm{S}}(\boldsymbol{U}_{h,1},q_1)=0,\end{cases}$$

and in Ω_2,

$$(W_{h,\mathrm{D}})\begin{cases}\text{Find } P_{h,2}\in M_2^h, \text{s.t.}\\ \forall q_2\in M_2^h,\ a_{\mathrm{D},h}(P_{h,2},q_2)=(f_2,q_2)_{\Omega_2}\\ +(\boldsymbol{U}_1^H\cdot\boldsymbol{n}_{12},q_2)_{\Gamma_{12}}+(g_{\mathrm{N}},q_2)_{\Gamma_{2\mathrm{N}}}\\ +\epsilon_2\sum_{e\in\Gamma_{2\mathrm{D}}}(\boldsymbol{K}\nabla q_2\cdot\boldsymbol{n}_e,g_{\mathrm{D}})_e+\sum_{e\in\Gamma_{2\mathrm{D}}}\frac{\sigma}{|e|}(g_{\mathrm{D}},q_2)_e.\end{cases}$$

Remark 1. For all schemes described above, the Darcy velocity is obtained by the following formula:

$\boldsymbol{U}_{h,2}=-\boldsymbol{K}\nabla P_{h,2}$.

4.3. Convergence result for the Stokes/Darcy problem

In this section, we obtain a priori error estimates for the numerical error of the two-grid approach. The main idea is to bound the difference of the solution of the coupled method obtained on the fine mesh and the solution of the two-grid method. For simplicity, we assume that only homogeneous Dirichlet boundary conditions are prescribed on the Darcy boundary Γ_2 and that the pressure test space is given by (16). We first need the following lemma.

Lemma 1. *Let $\boldsymbol{U}_1^h\in\boldsymbol{X}_1^h, P_1^h\in M_1^h, P_2^h\in M_2^h$ be the solution of the fully coupled Stokes–Darcy problem $\left(W_S^h\right)$ on a mesh of size h. Assume that $\epsilon_2=-1$ and that the penalty parameter σ_2 is large enough. Then, there is a constant C independent of h such that the following error bounds hold*

$$\|\boldsymbol{D}\left(\boldsymbol{u}_1-\boldsymbol{U}_1^h\right)\|_{L^2(\Omega_1)}+\|p_1-P_1^h\|_{L^2(\Omega_1)}+\|p_2-P_2^h\|_{\mathrm{DG}}\leqslant Ch, \tag{17}$$

$$\|\boldsymbol{u}_1-\boldsymbol{U}_1^h\|_{L^2(\Omega_1)}+\|p_2-P_2^h\|_{L^2(\Omega_2)}\leqslant Ch^2. \tag{18}$$

Proof. The proof of (17) can be obtained by following closely the arguments given in [1,4]. For the bound (18), an Aubin–Nitsche technique is employed. For this, it is assumed that the solution of the dual problem is regular enough. More precisely, if $\boldsymbol{f_1}\in L^2(\Omega_1)^2$ and $f_2\in L^2(\Omega_2)$, then the weak solution $(\boldsymbol{u_1},p_1,p_2)$ belongs to $H^2(\Omega_1)^2\times H^1(\Omega_1)\times H^2(\Omega_2)$. Details of the proof can be found in [32]. □

Remark 2. In the case $\epsilon_2=1$, the bound (17) still holds. However, the Aubin–Nitsche lift cannot be applied and the bound (18) is not valid. Using Sobolev's inequalities combined with Korn's inequality, we can obtain a suboptimal rate $\mathcal{O}(h)$ for the error in Stokes velocity and Darcy pressure in the L^2 norm.

Theorem 2. *Let $\boldsymbol{U}_{h,1}\in\boldsymbol{X}_1^h, P_{h,1}\in M_1^h, P_{h,2}\in M_2^h$ be the solution of $(W_{h,SD})$. Assume that $\epsilon_2=-1$ and that the penalty parameter σ_2 is large enough. There is a constant C independent of H and h such that*

$$\|\boldsymbol{D}(\boldsymbol{u}_1-\boldsymbol{U}_{h,1})\|_{L^2(\Omega_1)}+\|p_1-P_{h,1}\|_{L^2(\Omega_1)}\leqslant C(h+H^{3/2}), \tag{19}$$

$$\|p_2-P_{h,2}\|_{\mathrm{DG}}\leqslant C(h+H^2+hH). \tag{20}$$

Proof. Using triangle inequality, we have

$$\|\boldsymbol{D}(\boldsymbol{u}_1-\boldsymbol{U}_{h,1})\|_{L^2(\Omega_1)}\leqslant\|\boldsymbol{D}(\boldsymbol{u}_1-\boldsymbol{U}_1^h)\|_{L^2(\Omega_1)}+\|\boldsymbol{D}(\boldsymbol{U}_1^h-\boldsymbol{U}_{h,1})\|_{L^2(\Omega_1)},$$
$$\|p_1-P_{h,1}\|_{L^2(\Omega_1)}\leqslant\|p_1-P_1^h\|_{L^2(\Omega_1)}+\|P_1^h-P_{h,1}\|_{L^2(\Omega_1)},$$
$$\|p_2-P_{h,2}\|_{\mathrm{DG}}\leqslant\|p_2-P_2^h\|_{\mathrm{DG}}+\|P_2^h-P_{h,2}\|_{\mathrm{DG}}.$$

Using the previous lemma, it suffices to bound the second terms in the right-hand side of the inequalities above. Taking the difference between the fully coupled scheme $\left(W_S^h\right)$ and the decoupled problem $(W_{h,SD})$, we have

$$\begin{aligned}\forall\boldsymbol{v}_1\in\boldsymbol{X}_1^h,\forall q_2\in M_2^h,\quad & a_{\mathrm{S}}(\boldsymbol{U}_{h,1}-\boldsymbol{U}_1^h,\boldsymbol{v}_1)\\ &+\frac{1}{G}\left(\left(\boldsymbol{U}_{h,1}-\boldsymbol{U}_1^h\right)\cdot\boldsymbol{\tau}_{12},\boldsymbol{v}_1\cdot\boldsymbol{\tau}_{12}\right)_{\Gamma_{12}}+b_{\mathrm{S}}\left(\boldsymbol{v}_1,P_{h,1}-P_1^h\right)\\ &+a_{\mathrm{D},h}\left(P_{h,2}-P_2^h,q_2\right)=\left(P_2^h-P_2^H,\boldsymbol{v}_1\cdot\boldsymbol{n}_{12}\right)_{\Gamma_{12}}\\ &+\left(\left(\boldsymbol{U}_1^H-\boldsymbol{U}_1^h\right)\cdot\boldsymbol{n}_{12},q_2\right)_{\Gamma_{12}}\end{aligned} \tag{21}$$

$$\forall q_1\in M_1^h,\quad b_{\mathrm{S}}\left(\boldsymbol{U}_{h,1}-\boldsymbol{U}_1^h,q_1\right)=0. \tag{22}$$

We choose first $v_1=\mathbf{0}$ and $q_2=P_{h,2}-P_2^h$ in (21). We recall that the bilinear form $a_{\mathrm{D},h}$ is coercive. If the symmetric version is chosen ($\epsilon_2=-1$), then the form is coercive if the penalty parameter is large enough [33]. Therefore, there is a constant C such that

$$C\|P_{h,2}-P_2^h\|_{\mathrm{DG}}^2\leqslant\left|\left(\left(\boldsymbol{U}_1^H-\boldsymbol{U}_1^h\right)\cdot\boldsymbol{n}_{12},P_{h,2}-P_2^h\right)_{\Gamma_{12}}\right|.$$

Note that in the remainder of the paper, C denotes a generic constant independent of the mesh sizes. In addition, since the restriction of $P_{h,2}-P_2^h$ on the interface belongs to $H_{00}^{1/2}(\Gamma_{12})$, we can define the harmonic extension $\varphi\in H^1(\Omega_1)$ satisfying

$$\begin{aligned}&-\Delta\varphi=0,\quad\text{in }\Omega_1,\\ &\varphi=P_{h,2}-P_2^h,\quad\text{on }\Gamma_{12},\\ &\varphi=0,\quad\text{on }\Gamma_1.\end{aligned}$$

Then the bound $\|\varphi\|_{H^1(\Omega_1)}\leqslant C\|P_{h,2}-P_2^h\|_{\mathrm{DG}}$ holds. Using Green's theorem, we have

$$\begin{aligned}\left(\left(\boldsymbol{U}_1^H-\boldsymbol{U}_1^h\right)\cdot\boldsymbol{n}_{12},P_{h,2}-P_2^h\right)_{\Gamma_{12}}&=\left(\nabla\cdot\left(\boldsymbol{U}_1^H-\boldsymbol{U}_1^h\right),\varphi\right)_{\Omega_1}\\ &\quad+\left(\boldsymbol{U}_1^H-\boldsymbol{U}_1^h,\nabla\varphi\right)_{\Omega_1}.\end{aligned}$$

Using the previous lemma and the fact that $b_S\left(\boldsymbol{U}_1^H - \boldsymbol{U}_1^h, q\right) = 0$ for any test function $q \in M_1^H$ we have

$$\begin{aligned} C\|P_{h,2} - P_2^h\|_{\mathrm{DG}}^2 &\leqslant \|\boldsymbol{U}_1^H - \boldsymbol{U}_1^h\|_{L^2(\Omega_1)}\|\varphi\|_{H^1(\Omega_1)} \\ &+ \inf_{q\in M_1^H}|(\nabla\cdot(\boldsymbol{U}_1^H - \boldsymbol{U}_1^h), \varphi - q)_{\Omega_1}| \leqslant C\|\boldsymbol{U}_1^H - \boldsymbol{U}_1^h\|_{L^2(\Omega_1)}\|P_{h,2} - P_2^h\|_{\mathrm{DG}} \\ &+ CH\|\nabla(\boldsymbol{U}_1^H - \boldsymbol{U}_1^h)\|_{L^2(\Omega_1)}\|\varphi\|_{H^1(\Omega_1)} \leqslant C(h^2 + H^2 + hH)\|P_{h,2} - P_2^h\|_{\mathrm{DG}}, \end{aligned}$$

which yields (20).

Next, we choose $\boldsymbol{v}_1 = \boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h$ and $q_2 = 0$ in (21) and $q_1 = P_{h,1} - P_1^h$ in (22). We have using triangle inequality and a Sobolev embedding

$$\begin{aligned} 2\nu\|\boldsymbol{D}(\boldsymbol{U}_{h,1} - \boldsymbol{u}_1^h)\|_{L^2(\Omega_1)}^2 &\leqslant \|P_2^h - P_2^H\|_{L^2(\Gamma_{12})}\|(\boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h)\cdot\boldsymbol{n}_{12}\|_{L^2(\Gamma_{12})} \\ &\leqslant C(\|p_2 - P_2^h\|_{L^2(\Gamma_{12})} \\ &\quad + \|p_2 - P_2^H\|_{L^2(\Gamma_{12})})\|\boldsymbol{D}(\boldsymbol{U}_{h,1} - \boldsymbol{u}_1^h)\|_{L^2(\Omega_1)}. \end{aligned}$$

From trace inequalities [1], we have

$$\|p_2 - P_2^h\|_{L^2(\Gamma_{12})} \leqslant Ch^{3/2}, \quad \|p_2 - P_2^H\|_{L^2(\Gamma_{12})} \leqslant CH^{3/2}. \tag{23}$$

Finally, using Young's inequality, we conclude:

$$\|\boldsymbol{D}(\boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h)\|_{L^2(\Omega_1)}^2 \leqslant C(h^3 + H^3). \tag{24}$$

The error bound for the Stokes pressure is obtained by using the inf-sup condition and (21):

$$\begin{aligned} \|P_1^h - P_{h,1}\|_{L^2(\Omega_1)} &\leqslant C \sup_{\boldsymbol{v}_1\in\boldsymbol{X}_1^h} \frac{b_S(\boldsymbol{v}_1, P_1^h - P_{h,1})}{\|\boldsymbol{D}(\boldsymbol{v}_1)\|_{L^2(\Omega_1)}} \\ &\leqslant C \sup_{\boldsymbol{v}_1\in\boldsymbol{X}_1^h} \frac{1}{\|\boldsymbol{D}(\boldsymbol{v}_1)\|_{L^2(\Omega_1)}}\Big(a_S(\boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h, \boldsymbol{v}_1) \\ &\quad + \frac{1}{G}\left(\left(\boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h\right)\cdot\boldsymbol{\tau}_{12}, \boldsymbol{v}_1\cdot\boldsymbol{\tau}_{12}\right)_{\Gamma_{12}} \\ &\quad + \left(P_2^H - P_2^h, \boldsymbol{v}_1\cdot\boldsymbol{n}_{12}\right)_{\Gamma_{12}}\Big) \\ &\leqslant C\left(\|\boldsymbol{D}(\boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h)\|_{L^2(\Omega_1)} + \|P_2^H - P_2^h\|_{L^2(\Gamma_{12})}\right). \end{aligned}$$

We conclude by using (23) and (24). □

Remark 3. The choice $h = H^2$ yields a first order method for the error in the Darcy pressure. The error estimate is suboptimal for the error in the Stokes velocity and pressure as it yields $\mathcal{O}(h^{3/4})$. The numerical results below show an improved rate $\mathcal{O}(h)$.

Remark 4. In the case $\epsilon_2 = 1$, one can check that the leading term in (19) and (20) are $\mathcal{O}(H)$. This yields a suboptimal theoretical rate $\mathcal{O}(h^{1/2})$ for the choice $h = H^2$. Numerically the observed rate remains $\mathcal{O}(h)$.

4.4. Convergence result for the Navier–Stokes/Darcy problem

The nonlinear term $\boldsymbol{u}_1\cdot\nabla\boldsymbol{u}_1$ in the momentum Eq. (4) makes the error analysis of $\left(W_{NS}^h\right)$ and $(W_{h,NSD})$ more difficult and technical. In this section, we only highlight the differences with respect to the linear case. As in the previous section, only homogeneous Dirichlet boundary conditions are prescribed on the Darcy boundary and the pressure test space is defined by (16). Existence of a solution and convergence of the scheme are obtained under a *small data condition* given below:

$$\nu^4 > C^*\left(\|\boldsymbol{f}_1\|_{L^2(\Omega_1)}^2 + \nu\|f_2\|_{L^2(\Omega_2)}^2\right). \tag{25}$$

The constant C^* depends on several constants from Sobolev embeddings, Korn's inequality and trace inequalities. This small data condition means that the Reynolds number or the source functions cannot be too large. Under this condition, a similar bound to (17) can be proved. In addition, under the assumption of regularity of the dual problem, the optimal bound (18) is valid. Next we observe that the proof for the error estimate (20) does not involve the momentum equation. Thus, this estimate is valid for the Navier–Stokes/Darcy problem with the same proof as for the Stokes/Darcy problem. The error bound for the Navier–Stokes velocity and pressure is similar than the bound (19). However the proof differs as one has to control the nonlinear terms. In this case, taking the difference between the fully coupled scheme $\left(W_{NS}^h\right)$ and the decoupled problem $(W_{h,NS})$ yields:

$$\begin{aligned} &\forall\boldsymbol{v}_1 \in \boldsymbol{X}_1^h, \quad a_S(\boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h, \boldsymbol{v}_1) + \frac{1}{G}((\boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h)\cdot\boldsymbol{\tau}_{12}, \boldsymbol{v}_1\cdot\boldsymbol{\tau}_{12})_{\Gamma_{12}} \\ &\quad + b_S\left(\boldsymbol{v}_1, P_{h,1} - P_1^h\right) = \left(P_2^h - P_2^H, \boldsymbol{v}_1\cdot\boldsymbol{n}_{12}\right)_{\Gamma_{12}} \\ &\quad + c_{NS}(\boldsymbol{U}_1^h; \boldsymbol{U}_1^h, \boldsymbol{v}_1) - c_{NS}\left(\boldsymbol{U}_1^H; \boldsymbol{U}_{h,1}, \boldsymbol{v}_1\right) \end{aligned} \tag{26}$$

$$\forall q_1 \in M_1^h, \quad b_S\left(\boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h, q_1\right) = 0. \tag{27}$$

Taking $\boldsymbol{v}_1 = \boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h$ and $q_1 = P_{h,1} - P_1^h$ in the equations above yields:

$$\begin{aligned} 2\nu\|\boldsymbol{D}(\boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h)\|_{L^2(\Omega_1)}^2 &\leqslant \|P_2^h - P_2^H\|_{L^2(\Gamma_{12})}\|(\boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h)\cdot\boldsymbol{n}_{12}\|_{L^2(\Gamma_{12})} \\ &\quad + |c_{NS}\left(\boldsymbol{U}_1^h; \boldsymbol{U}_1^h, \boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h\right) \\ &\quad - c_{NS}\left(\boldsymbol{U}_1^H; \boldsymbol{U}_{h,1}, \boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h\right)|. \end{aligned}$$

The first term is handled as for the linear case. The second term is rewritten as:

$$\begin{aligned} &c_{NS}\left(\boldsymbol{U}_1^h; \boldsymbol{U}_1^h, \boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h\right) - c_{NS}\left(\boldsymbol{U}_1^H; \boldsymbol{U}_{h,1}, \boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h\right) \\ &\quad = c_{NS}\left(\boldsymbol{U}_1^h - \boldsymbol{U}_1^H; \boldsymbol{U}_1^h, \boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h\right) - c_{NS}\left(\boldsymbol{U}_1^H; \boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h, \boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h\right). \end{aligned}$$

Using Sobolev's embeddings and trace inequalities, we have for two constants C and $\widetilde{C}$:

$$\begin{aligned} &c_{NS}(\boldsymbol{U}_1^h - \boldsymbol{U}_1^H; \boldsymbol{U}_1^h, \boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h) \\ &\quad \leqslant C\|\boldsymbol{D}(\boldsymbol{U}_1^h - \boldsymbol{U}_1^H)\|_{L^2(\Omega_1)}\|\boldsymbol{D}(\boldsymbol{U}_1^h)\|_{L^2(\Omega_1)}\|\boldsymbol{D}(\boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h)\|_{L^2(\Omega_1)} \\ &\quad \leqslant C(h + H)\|\boldsymbol{D}(\boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h)\|_{L^2(\Omega_1)} \end{aligned}$$

and

$$\begin{aligned} c_{NS}\left(\boldsymbol{U}_1^H; \boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h, \boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h\right) &\leqslant C\|\boldsymbol{D}\left(\boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h\right)\|_{L^2(\Omega_1)}^2\|\boldsymbol{D}\left(\boldsymbol{U}_1^H\right)\|_{L^2(\Omega_1)} \\ &\leqslant \widetilde{C}\|\boldsymbol{D}\left(\boldsymbol{U}_{h,1} - \boldsymbol{U}_1^h\right)\|_{L^2(\Omega_1)}^2. \end{aligned}$$

Note that we have used the fact that $\|\boldsymbol{D}\left(\boldsymbol{U}_1^h\right)\|_{L^2(\Omega_1)}$ and $\|\boldsymbol{D}\left(\boldsymbol{U}_1^H\right)\|_{L^2(\Omega_1)}$ are bounded by constants that depend on the data $\nu, \boldsymbol{f}_1$ and f_2. By carefully defining the constant C^* in the small data condition, we can then conclude. Remarks 2–4 also hold true.

5. Numerical examples

5.1. Convergence study for Stokes and Darcy problem

We compute numerical convergence rates for the two-grid algorithm applied to the Stokes/Darcy problem. We also compare the accuracy of the two-grid solution with the solution obtained

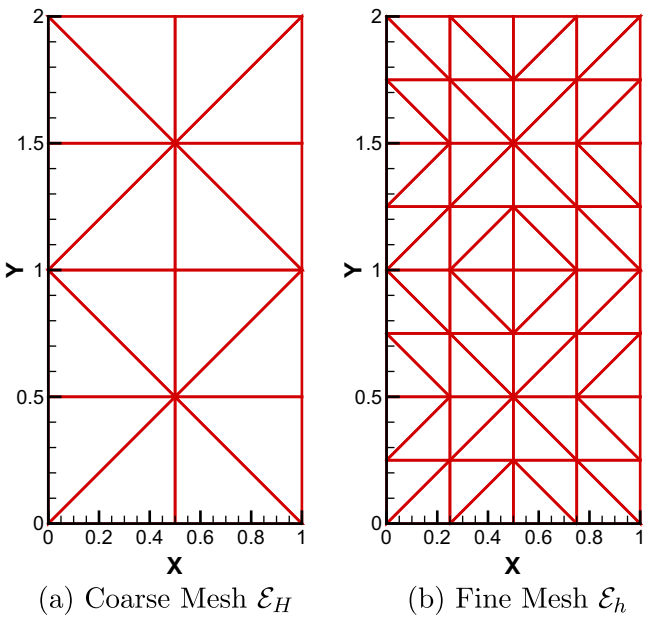

Fig. 1. Computational meshes $H = 0.5$, $h = 0.25$.

by solving the fully coupled problem. The computational domain $\Omega \subset \mathbb{R}^2$ is subdivided into $\Omega_1 = (0,1) \times (0,1)$ and $\Omega_2 = (0,1) \times (1,2)$. For each given mesh of size h, we solve the fully coupled problem on a coarse mesh of size $H = \sqrt{h}$ and the decoupled problems on a fine mesh of size h. An example of such meshes is given in Fig. 1. The boundary conditions are chosen in such a way that the exact solution to the coupled problem is:

$$\begin{aligned}\boldsymbol{u}_1(x,y) &= (y^2 - 2y + 2x, x^2 - x - 2y), \quad p_1(x,y) \\ &= x^2y + xy + y^2 - 4.0, \quad p_2(x,y) = -x^2y + xy + y^2.\end{aligned}$$

For simplicity we set $\boldsymbol{K} = \boldsymbol{I}$ and $\nu = 1$. First we solve the fully coupled problem on a sequence of fine meshes of size h. For the subsurface part, the NIPG method is used with penalty $\sigma = 1$ and with order of approximation $k_2 = 1$. Table 1 shows the errors and convergence rates for the fully coupled problem. As expected from the theory [4] we observe order one convergence rate in the Stokes and Darcy velocities in the energy norm. The Stokes pressure has a slightly higher convergence rate than the expected order one. This phenomenom was also observed in [13]. We also show the rates for the Stokes velocity and Darcy pressure in the L^2 norm. We recover the optimal rate of two.

We then solve problem $(W_{h,\mathrm{SD}})$ on the same sequence of meshes as in Table 1. Tables 2 and 3 show the errors and convergence rates for $(W_{h,\mathrm{SD}})$ using the NIPG method with $\sigma = 1$ and the SIPG method with $\sigma = 6$ in Ω_2 respectively. First we note that the errors obtained from the NIPG and SIPG solutions are comparable. For the SIPG method the expected convergence rate is obtained by making the penalty parameter large enough. This is due to the fact that the coercivity of the bilinear form a_{D} for the SIPG method depends on the choice of the penalty parameter. From [33], we choose $\sigma = k_2(k_2 + 1)\cot(\theta)$, where θ is the smallest angle over all triangles in the mesh. We obtain order one convergence for the errors in the energy norm in the parameter h for both methods. The order of convergence is better than the predicted theoretical convergence rate of $h^{3/4}$. The errors are comparable to those obtained by solving the fully coupled problem (Table 1) for both methods. However the Stokes pressure error is relatively larger for the two-grid problem compared to the fully coupled problem.

Next we increase the degree of approximation in Ω_2 to two. Tables 4 and 5 show the errors and convergence rates for the two-grid algorithm using the NIPG method with $\sigma = 0$ and SIPG method with $\sigma = 18$ in Ω_2 respectively. Increasing the degree of approximation in Ω_2 improves the quality of the Darcy pressure solution. For example on a mesh of size $\frac{1}{64}$ the error in the Darcy velocity in Table 4 is about seven times smaller compared to Tables 2 and 3. The Stokes velocity error is unchanged from the case $k_2 = 1$ to $k_2 = 2$. The Stokes pressure error decreases by a factor of four.

Further, we increase the degree of approximation in Ω_2 to three. Table 6 shows the errors and convergence rates for this case. We note that the order of approximation remains one in the Stokes domain and increasing the order to three does not have any further improvement in the accuracy of the solution in both domains. This is expected because the order of approximation in Ω_1 remains one.

It is clear that the two-grid method performs well for the coupled Stokes and Darcy problem. The real advantage of this method will be shown in the case of the coupled Navier–Stokes and Darcy problem because we avoid solving a non-linear problem on the fine mesh.

5.2. Convergence study for Navier–Stokes and Darcy problem

In this section we compare the accuracy of the two-grid method for the Navier–Stokes/Darcy problem with the accuracy of the fully coupled method. The problem set-up and input data are the same

Table 1
Errors and rates for $\left(W_S^h\right)$(FEM/NIPG$\sigma = 1, k_2 = 1$).

h	$\|\boldsymbol{U}_1^h - \boldsymbol{u}_1\|_{L^2(\Omega_1)}$	$\|P_1^h - p_1\|_{L^2(\Omega_1)}$	$\|\boldsymbol{D}(\boldsymbol{U}_1^h - \boldsymbol{u}_1)\|_{L^2(\Omega_1)}$	$\|P_2^h - p_2\|_{L^2(\Omega_2)}$	$\|\nabla(P_2^h - p_2)\|_{L^2(\Omega_2)}$
$\frac{1}{4}$	1.591×10^{-2}	6.346×10^{-2}	1.365×10^{-1}	1.341×10^{-2}	1.637×10^{-1}
$\frac{1}{16}$	9.984×10^{-4}	6.645×10^{-3}	3.390×10^{-2}	8.210×10^{-4}	3.856×10^{-2}
$\frac{1}{64}$	6.227×10^{-5}	7.363×10^{-4}	8.469×10^{-3}	5.150×10^{-5}	9.552×10^{-3}
Rate (h)	2.00	1.58	1.00	2.00	1.00

Table 2
Errors and rates for $(W_{h,\mathrm{SD}})$ (FEM/NIPG $\sigma = 1, k_2 = 1$).

h	H	$\|\boldsymbol{U}_{h,1} - \boldsymbol{u}_1\|_{L^2(\Omega_1)}$	$\|P_{h,1} - p_1\|_{L^2(\Omega_1)}$	$\|\boldsymbol{D}(\boldsymbol{U}_{h,1} - \boldsymbol{u}_1)\|_{L^2(\Omega_1)}$	$\|P_{h,2} - p_2\|_{L^2(\Omega_2)}$	$\|\nabla(P_{h,2} - p_2)\|_{L^2(\Omega_2)}$
$\frac{1}{4}$	$\frac{1}{2}$	1.599×10^{-2}	6.647×10^{-2}	1.364×10^{-1}	1.235×10^{-2}	1.657×10^{-1}
$\frac{1}{16}$	$\frac{1}{4}$	9.984×10^{-4}	1.571×10^{-2}	3.397×10^{-2}	1.018×10^{-3}	3.923×10^{-2}
$\frac{1}{64}$	$\frac{1}{8}$	6.073×10^{-5}	4.349×10^{-3}	8.473×10^{-3}	2.699×10^{-4}	9.638×10^{-3}
Rate (h)	–	2.00	0.92	1.00	0.95	1.00

Table 3
Errors and rates for ($W_{h,\mathrm{SD}}$) (FEM/SIPG $\sigma = 6, k_2 = 1$).

h	H	$\|\boldsymbol{U}_{h,1} - \boldsymbol{u}_1\|_{L^2(\Omega_1)}$	$\|P_{h,1} - p_1\|_{L^2(\Omega_1)}$	$\|\boldsymbol{D}(\boldsymbol{U}_{h,1} - \boldsymbol{u}_1)\|_{L^2(\Omega_1)}$	$\|P_{h,2} - p_2\|_{L^2(\Omega_2)}$	$\|\nabla(P_{h,2} - p_2)\|_{L^2(\Omega_2)}$
$\frac{1}{4}$	$\frac{1}{2}$	1.594×10^{-2}	7.383×10^{-2}	1.363×10^{-1}	8.788×10^{-3}	1.703×10^{-1}
$\frac{1}{16}$	$\frac{1}{4}$	1.002×10^{-3}	1.489×10^{-2}	3.397×10^{-2}	1.471×10^{-3}	4.137×10^{-2}
$\frac{1}{64}$	$\frac{1}{8}$	6.616×10^{-5}	4.166×10^{-3}	8.473×10^{-3}	3.115×10^{-4}	1.011×10^{-2}
Rate (h)	–	2.00	0.92	1.00	1.12	1.00

Table 4
Errors and rates for ($W_{h,\mathrm{SD}}$) (FEM/NIPG $\sigma = 0, k_2 = 2$).

h	H	$\|\boldsymbol{U}_{h,1} - \boldsymbol{u}_1\|_{L^2(\Omega_1)}$	$\|P_{h,1} - p_1\|_{L^2(\Omega_1)}$	$\|\boldsymbol{D}(\boldsymbol{U}_{h,1} - \boldsymbol{u}_1)\|_{L^2(\Omega_1)}$	$\|P_{h,2} - p_2\|_{L^2(\Omega_2)}$	$\|\nabla(P_{h,2} - p_2)\|_{L^2(\Omega_2)}$
$\frac{1}{4}$	$\frac{1}{2}$	1.599×10^{-2}	6.577×10^{-2}	1.363×10^{-1}	5.134×10^{-3}	2.556×10^{-2}
$\frac{1}{16}$	$\frac{1}{4}$	9.928×10^{-4}	7.708×10^{-3}	3.396×10^{-2}	1.213×10^{-3}	5.654×10^{-3}
$\frac{1}{64}$	$\frac{1}{8}$	6.258×10^{-5}	1.304×10^{-3}	8.470×10^{-3}	2.981×10^{-4}	1.378×10^{-3}
Rate (h)	–	2.00	1.28	1.00	1.00	1.00

Table 5
Errors and rates for ($W_{h,\mathrm{SD}}$) (FEM/SIPG $\sigma = 18, k_2 = 2$).

h	H	$\|\boldsymbol{U}_{h,1} - \boldsymbol{u}_1\|_{L^2(\Omega_1)}$	$\|P_{h,1} - p_1\|_{L^2(\Omega_1)}$	$\|\boldsymbol{D}(\boldsymbol{U}_{h,1} - \boldsymbol{u}_1)\|_{L^2(\Omega_1)}$	$\|P_{h,2} - p_2\|_{L^2(\Omega_2)}$	$\|\nabla(P_{h,2} - p_2)\|_{L^2(\Omega_2)}$
$\frac{1}{4}$	$\frac{1}{2}$	1.599×10^{-2}	6.470×10^{-2}	1.364×10^{-1}	4.965×10^{-3}	2.470×10^{-2}
$\frac{1}{16}$	$\frac{1}{4}$	9.927×10^{-4}	7.253×10^{-3}	3.395×10^{-2}	1.192×10^{-3}	5.585×10^{-3}
$\frac{1}{64}$	$\frac{1}{8}$	6.236×10^{-5}	1.124×10^{-3}	8.473×10^{-3}	2.968×10^{-4}	1.374×10^{-3}
Rate (h)	–	2.00	1.34	1.00	1.00	1.00

Table 6
Errors and rates for ($W_{h,\mathrm{SD}}$) (FEM/NIPG $\sigma = 0, k_2 = 3$).

h	H	$\|\boldsymbol{U}_{h,1} - \boldsymbol{u}_1\|_{L^2(\Omega_1)}$	$\|P_{h,1} - p_1\|_{L^2(\Omega_1)}$	$\|\boldsymbol{D}(\boldsymbol{U}_{h,1} - \boldsymbol{u}_1)\|_{L^2(\Omega_1)}$	$\|P_{h,2} - p_2\|_{L^2(\Omega_2)}$	$\|\nabla(P_{h,2} - p_2)\|_{L^2(\Omega_2)}$
$\frac{1}{4}$	$\frac{1}{2}$	1.599×10^{-2}	6.417×10^{-2}	1.363×10^{-1}	4.942×10^{-3}	2.328×10^{-2}
$\frac{1}{16}$	$\frac{1}{4}$	9.926×10^{-4}	7.196×10^{-3}	3.395×10^{-2}	1.191×10^{-3}	5.579×10^{-3}
$\frac{1}{64}$	$\frac{1}{8}$	6.233×10^{-5}	1.119×10^{-3}	8.470×10^{-3}	2.967×10^{-4}	1.375×10^{-3}
Rate (h)	–	2.00	1.34	1.00	1.00	1.00

as in the previous section. Table 7 shows the errors and convergence rates for the solution from the fully coupled Navier–Stokes and Darcy problem. The tolerance for stopping the Picard iterations is equal to 10^{-7}. We observe order one convergence rate in the energy norm for the Navier–Stokes velocity and Darcy pressure. We then compare the solution from the fully coupled problem to the decoupled Navier–Stokes and Darcy problems ($W_{h,\mathrm{NSD}}$). From Table 8 we note that the errors from the two-grid method are compa-

Table 7
Errors and rates for $\left(W^h_{\mathrm{NS}}\right)$(FEM/NIPG $\sigma = 1, k_2 = 1$).

h	$\|\boldsymbol{U}^h_1 - \boldsymbol{u}_1\|_{L^2(\Omega_1)}$	$\|P^h_1 - p_1\|_{L^2(\Omega_1)}$	$\|\boldsymbol{D}(\boldsymbol{U}^h_1 - \boldsymbol{u}_1)\|_{L^2(\Omega_1)}$	$\|P^h_2 - p_2\|_{L^2(\Omega_2)}$	$\|\nabla(P^h_2 - p_2)\|_{L^2(\Omega_2)}$
$\frac{1}{4}$	1.598×10^{-2}	6.889×10^{-2}	1.366×10^{-1}	2.598×10^{-2}	1.625×10^{-1}
$\frac{1}{16}$	9.988×10^{-4}	6.754×10^{-3}	3.390×10^{-2}	1.592×10^{-3}	3.841×10^{-2}
$\frac{1}{64}$	6.232×10^{-5}	7.403×10^{-4}	8.467×10^{-3}	9.948×10^{-5}	9.541×10^{-3}
Rate (h)	2.00	1.59	1.00	2.00	1.00

Table 8
Errors and rates for ($W_{h,\mathrm{NSD}}$) (FEM/NIPG $\sigma = 1, k_2 = 1$).

h	H	$\|\boldsymbol{U}_{h,1} - \boldsymbol{u}_1\|_{L^2(\Omega_1)}$	$\|P_{h,1} - p_1\|_{L^2(\Omega_1)}$	$\|\boldsymbol{D}(\boldsymbol{U}_{h,1} - \boldsymbol{u}_1)\|_{L^2(\Omega_1)}$	$\|P_{h,2} - p_2\|_{L^2(\Omega_2)}$	$\|\nabla(P_{h,2} - p_2)\|_{L^2(\Omega_2)}$
$\frac{1}{4}$	$\frac{1}{2}$	1.594×10^{-2}	9.7277×10^{-2}	1.365×10^{-1}	4.258×10^{-2}	1.708×10^{-1}
$\frac{1}{16}$	$\frac{1}{4}$	9.696×10^{-4}	1.728×10^{-2}	3.399×10^{-2}	7.068×10^{-3}	4.029×10^{-2}
$\frac{1}{64}$	$\frac{1}{8}$	7.137×10^{-5}	3.961×10^{-3}	8.477×10^{-3}	1.567×10^{-3}	9.896×10^{-3}
Rate (h)	–	2.00	1.00	1.00	1.20	1.20

rable to the fully coupled problem for the Navier–Stokes and Darcy velocities in Table 7. The convergence rates are order one for the velocity and pressure in both domains. This is better than the predicted order $h^{3/4}$ for the case $H = \sqrt{h}$. We note that the same improved rate has been reported in [22,23]. Next, we increase the order of approximation in Ω_2 to $k_2 = 2$. From Table 9 as expected the Darcy solution is more accurate compared to the order one approximation, the convergence rate in the velocity increases to two. In Table 10 we observe that the two-grid solution with $k_2 = 2$ is more accurate compared to the one with $k_2 = 1$ in Table 8. The Darcy velocity error decreases by a factor of almost 4, this in-turn results in an improvement in the Navier–Stokes velocity and pressure solutions in Ω_1. However, due to the fact that the order of approximation in Ω_1 is one only, the errors in Table 10 are larger compared to the coupled problem in Table 9. The convergence rate for the Navier–Stokes velocity and pressure in Ω_1 is order one, the same holds true for the Darcy velocity in Ω_2. We also present results for the SIPG method below. As we noted in the case with the Stokes and Darcy coupling the penalty parameter σ has to be chosen to be large enough in order to observe the expected convergence rate. Table 11 shows the errors and convergence rate for the fully coupled problem with $k_2 = 1$ using the SIPG method in Ω_2. Table 12 shows the errors and convergence rate from the two-grid method on the same meshes. We observe the same trend as in the case with the NIPG method above. The errors for the two-grid algorithm converge with order one and are comparable to the fully coupled problem.

Tables 13 and 14 show the errors and convergence rates for the fully coupled Navier–Stokes and Darcy model and the two-grid method respectively if quadratic polynomials are used in the

Table 9
Errors and rates for $\left(W_{\mathrm{NS}}^h\right)$(FEM/NIPG $\sigma = 0, k_2 = 2$).

h	$\|\boldsymbol{U}_1^h - \boldsymbol{u}_1\|_{L^2(\Omega_1)}$	$\|P_1^h - p_1\|_{L^2(\Omega_1)}$	$\|\boldsymbol{D}(\boldsymbol{U}_1^h - \boldsymbol{u}_1)\|_{L^2(\Omega_1)}$	$\|P_2^h - p_2\|_{L^2(\Omega_2)}$	$\|\nabla(P_2^h - p_2)\|_{L^2(\Omega_2)}$
$\frac{1}{4}$	1.599×10^{-2}	6.687×10^{-2}	1.366×10^{-1}	7.449×10^{-3}	1.463×10^{-2}
$\frac{1}{16}$	9.994×10^{-4}	6.685×10^{-3}	3.390×10^{-2}	4.953×10^{-4}	9.275×10^{-4}
$\frac{1}{64}$	6.232×10^{-5}	7.379×10^{-4}	8.467×10^{-3}	3.1225×10^{-5}	5.812×10^{-5}
Rate (h)	2.00	1.59	1.00	2.00	2.00

Table 10
Errors and rates for ($W_{h,\mathrm{NSD}}$) (FEM/NIPG $\sigma = 0, k_2 = 2$).

h	H	$\|\boldsymbol{U}_{h,1} - \boldsymbol{u}_1\|_{L^2(\Omega_1)}$	$\|P_{h,1} - p_1\|_{L^2(\Omega_1)}$	$\|\boldsymbol{D}(\boldsymbol{U}_{h,1} - \boldsymbol{u}_1)\|_{L^2(\Omega_1)}$	$\|P_{h,2} - p_2\|_{L^2(\Omega_2)}$	$\|\nabla(P_{h,2} - p_2)\|_{L^2(\Omega_2)}$
$\frac{1}{4}$	$\frac{1}{2}$	1.594×10^{-2}	7.758×10^{-2}	1.364×10^{-1}	2.562×10^{-2}	1.708×10^{-1}
$\frac{1}{16}$	$\frac{1}{4}$	9.674×10^{-4}	1.278×10^{-2}	3.397×10^{-2}	6.131×10^{-3}	1.057×10^{-2}
$\frac{1}{64}$	$\frac{1}{8}$	6.698×10^{-5}	2.882×10^{-3}	8.474×10^{-3}	1.511×10^{-5}	2.616×10^{-3}
Rate (h)	–	2.00	1.00	1.00	1.20	1.20

Table 11
Errors and rates for $\left(W_{\mathrm{NS}}^h\right)$(FEM/SIPG $\sigma = 6, k_2 = 1$).

h	$\|\boldsymbol{U}_1^h - \boldsymbol{u}_1\|_{L^2(\Omega_1)}$	$\|P_1^h - p_1\|_{L^2(\Omega_1)}$	$\|\boldsymbol{D}(\boldsymbol{U}_1^h - \boldsymbol{u}_1)\|_{L^2(\Omega_1)}$	$\|P_2^h - p_2\|_{L^2(\Omega_2)}$	$\|\nabla(P_2^h - p_2)\|_{L^2(\Omega_2)}$
$\frac{1}{4}$	1.600×10^{-2}	6.681×10^{-2}	1.366×10^{-1}	5.060×10^{-2}	1.612×10^{-1}
$\frac{1}{16}$	1.000×10^{-3}	6.682×10^{-3}	3.390×10^{-2}	4.000×10^{-4}	3.9791×10^{-2}
$\frac{1}{64}$	6.232×10^{-5}	7.378×10^{-4}	8.467×10^{-3}	2.618×10^{-5}	9.974×10^{-3}
Rate (h)	2.00	1.59	1.00	2.00	1.00

Table 12
Errors and rates for ($W_{h,\mathrm{NSD}}$) (FEM/SIPG $\sigma = 6, k_2 = 1$).

h	H	$\|\boldsymbol{U}_{h,1} - \boldsymbol{u}_1\|_{L^2(\Omega_1)}$	$\|P_{h,1} - p_1\|_{L^2(\Omega_1)}$	$\|\boldsymbol{D}(\boldsymbol{U}_{h,1} - \boldsymbol{u}_1)\|_{L^2(\Omega_1)}$	$\|P_{h,2} - p_2\|_{L^2(\Omega_2)}$	$\|\nabla(P_{h,2} - p_2)\|_{L^2(\Omega_2)}$
$\frac{1}{4}$	$\frac{1}{2}$	1.594×10^{-2}	7.931×10^{-2}	1.364×10^{-1}	5.289×10^{-2}	1.661×10^{-1}
$\frac{1}{16}$	$\frac{1}{4}$	9.747×10^{-4}	1.278×10^{-2}	3.397×10^{-2}	5.714×10^{-3}	4.153×10^{-2}
$\frac{1}{64}$	$\frac{1}{8}$	6.350×10^{-5}	2.842×10^{-3}	8.474×10^{-3}	1.482×10^{-3}	1.029×10^{-2}
Rate (h)	–	2.00	1.00	1.00	1.20	1.20

Table 13
Errors and rates for $\left(W_{\mathrm{NS}}^h\right)$(FEM/SIPG $\sigma = 18, k_2 = 2$).

h	$\|\boldsymbol{U}_1^h - \boldsymbol{u}_1\|_{L^2(\Omega_1)}$	$\|P_1^h - p_1\|_{L^2(\Omega_1)}$	$\|\boldsymbol{D}(\boldsymbol{U}_1^h - \boldsymbol{u}_1)\|_{L^2(\Omega_1)}$	$\|P_2^h - p_2\|_{L^2(\Omega_2)}$	$\|\nabla(P_2^h - p_2)\|_{L^2(\Omega_2)}$
$\frac{1}{4}$	1.599×10^{-2}	6.604×10^{-2}	1.366×10^{-1}	6.021×10^{-3}	1.301×10^{-2}
$\frac{1}{16}$	9.993×10^{-4}	6.668×10^{-3}	3.390×10^{-2}	3.759×10^{-4}	8.183×10^{-4}
$\frac{1}{64}$	6.233×10^{-5}	7.373×10^{-4}	8.469×10^{-3}	2.349×10^{-5}	5.123×10^{-5}
Rate (h)	2.00	1.59	1.00	2.00	1.00

Table 14
Errors and rates for ($W_{h,\mathrm{NSD}}$) (FEM/SIPG $\sigma = 18, k_2 = 2$).

h	H	$\|\boldsymbol{U}_{h,1} - \boldsymbol{u}_1\|_{L^2(\Omega_1)}$	$\|P_{h,1} - p_1\|_{L^2(\Omega_1)}$	$\|\boldsymbol{D}(\boldsymbol{U}_{h,1} - \boldsymbol{u}_1)\|_{L^2(\Omega_1)}$	$\|P_{h,2} - p_2\|_{L^2(\Omega_2)}$	$\|\nabla(P_{h,2} - p_2)\|_{L^2(\Omega_2)}$
$\frac{1}{4}$	$\frac{1}{2}$	1.594×10^{-2}	7.722×10^{-2}	1.364×10^{-1}	2.406×10^{-2}	4.206×10^{-2}
$\frac{1}{16}$	$\frac{1}{4}$	9.671×10^{-4}	1.249×10^{-2}	3.397×10^{-2}	6.014×10^{-3}	1.042×10^{-2}
$\frac{1}{64}$	$\frac{1}{8}$	6.764×10^{-5}	2.784×10^{-3}	8.477×10^{-3}	1.503×10^{-3}	2.606×10^{-3}
Rate (h)	–	2.00	1.00	1.00	1.20	1.20

subsurface. Increasing the order of approximation in Ω_2 to $k_2 = 2$ for the decoupled problem ($W_{h,\mathrm{NSD}}$) yields a more accurate Darcy velocity. However, if piecewise quadratics are used in the Darcy domain, we note that the errors in the Darcy pressure obtained with the fully coupled scheme are smaller than those obtained by the two-grid approach. The accuracy of the coarse solution is not enough to compensate for the errors introduced by decoupling the problems.

Figs. 2 and 3 are plots of the x and y components of the velocity solutions respectively for the fully coupled Navier–Stokes/Darcy scheme and the two-grid scheme. It is clear that even for the choice of mesh sizes $(h,H) = (0.25, 0.5)$, the two-grid solution gives the same information for the velocity than in the case of the fully coupled solution. Obtaining an accurate velocity field is important as it becomes input data for the contaminant transport problem [34].

5.3. Comparison of CPU times for the Navier–Stokes and Darcy problem

The advantage of the two-grid method for the Navier–Stokes and Darcy coupling is that it results in a linearized system. This means that on the fine grid we solve once the linear system resulting from an Oseen problem in Ω_1. This yields significant gains in computational time and efficiency of the solver.

Table 15 shows the CPU times for solving one linear system for the fully coupled Navier–Stokes/Darcy model and for the decoupled problems. It is clear that the reduced problems are faster to solve. We note that the linear systems are solved by a direct solver.

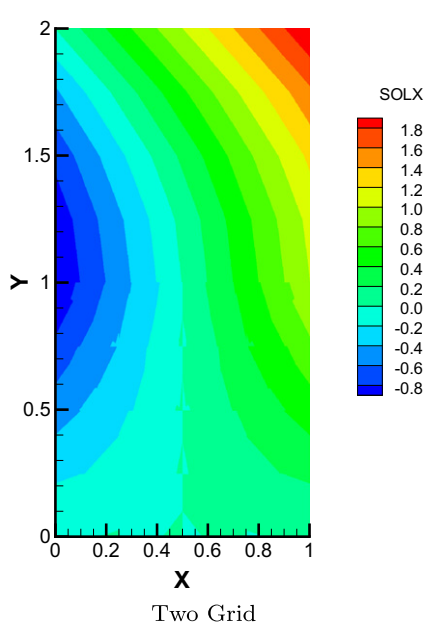

Fig. 2. x-Component of velocity $H = 0.5$, $h = 0.25$, *FEM/NIPG*, $k_2 = 2$, $\sigma = 0$.

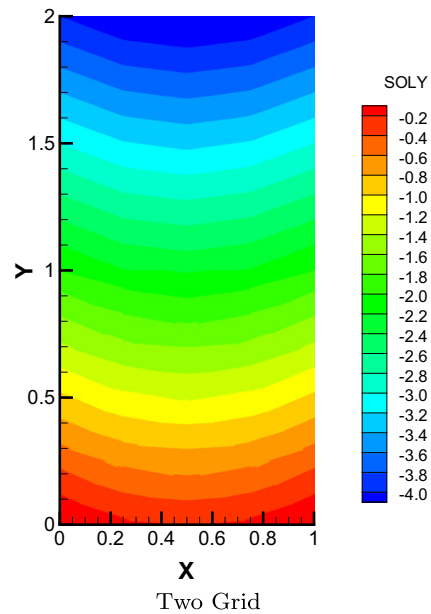

Fig. 3. y-Component of velocity $H = 0.5$, $h = 0.25$, *FEM/NIPG*, $k_2 = 2$, $\sigma = 0$.

Table 15
CPU times for one linear solve: $W^h_{\mathrm{NS}}, W_{h,NS}, W_{h,D}$, (*FEM/NIPG* $\sigma = 0, k_2 = 2$).

h	H	Fully Coupled $\left(t_{W^h_{\mathrm{NS}}}\right)$	Navier–Stokes $(t_{W_{h,\mathrm{NS}}})$	Darcy $(t_{W_{h,D}})$
$\frac{1}{16}$	$\frac{1}{4}$	1.29×10^0 s	6.00×10^{-2} s	1.20×10^{-1} s
$\frac{1}{64}$	$\frac{1}{8}$	$8.731 \times 10^{+1}$ s	1.69×10^0 s	7.36×10^0 s

Another important contribution to CPU time is the time spent in the assembly of the matrices for the linear system. In addition there is accumulation of time over all Picard iterations for the coupled problem. Table 16 gives the total time taken to assemble the linear system and to have the Picard iterations to converge. In this problem, the coupled problem on both coarse and fine meshes requires eight Picard iterations for a tolerance equal to 10^{-6}. From the table, we observe that the sequential CPU time for the two-grid approach is the sum $t_{W^h_{\mathrm{NS}}} + t_{W_{h,\mathrm{NS}}} + t_{W_{h,D}}$. For a mesh of size $h = 1/16$, this time is about 9 times larger than the CPU time for the fully coupled approach. If the mesh is refined ($h = 1/64$), the two-grid method is about 4 times faster than the fully coupled method.

In that case, we note that the linear system for the fully coupled problem is of size $(77{,}825 \times 77{,}825)$ whereas the decoupled problems are of size $(28{,}673 \times 28{,}673)$ and $(49{,}152 \times 49{,}152)$ for the Oseen and Darcy problems respectively.

Asymptotically, as the mesh size tends to zero, the computational time for the coarse grid solve will be much smaller than that of the decoupled problems. The gain of the two-grid approach is then due to the fact that only two linear reduced systems are solved.

Table 16
Total CPU times: $W^h_{NS}, W_{h,NS}, W_{h,D}$, ($FEM/NIPG$ $\sigma = 0, k_2 = 2$).

h	H	Fully coupled $\left(t_{W^h_{NS}}\right)$	Fully coupled $\left(t_{W^H_{NS}}\right)$	Navier–Stokes $(t_{W_{h,NS}})$	Darcy $(t_{W_{h,D}})$
$\frac{1}{16}$	$\frac{1}{4}$	9.4×10^0 s	3.2×10^{-1} s	6.0×10^{-1} s	1.6×10^{-1} s
$\frac{1}{64}$	$\frac{1}{8}$	$1.0 \times 10^{+3}$ s	1.1×10^0 s	2.4×10^2 s	1.1×10^1 s

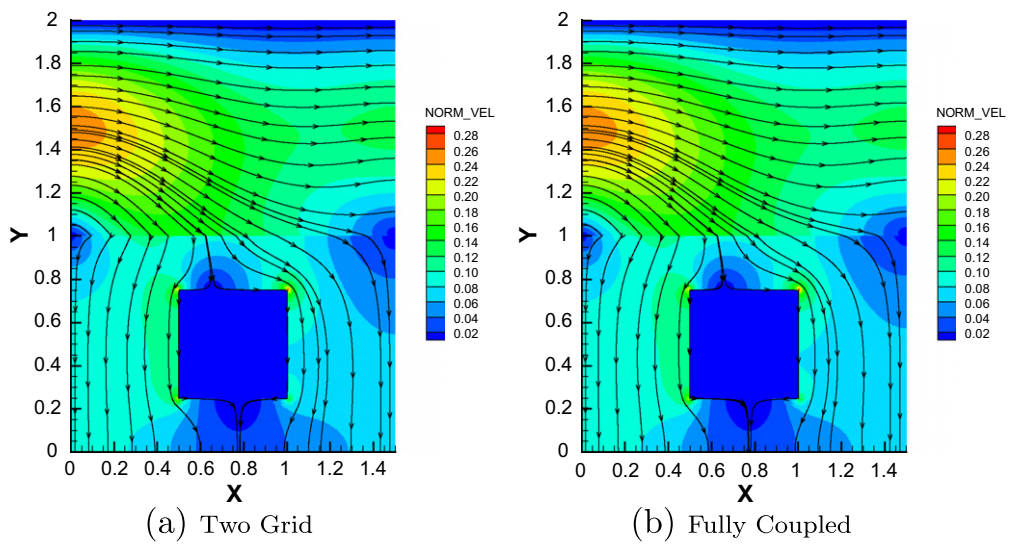

Fig. 4. Euclidian norm of velocity $H = \frac{1}{8}, h = \frac{1}{32}, FEM/NIPG, k_2 = 2, \sigma = 0$.

5.4. Case of impermeable inclusion in subsurface

Next we consider a case for which the exact solution for the coupled flow problem is unknown. The computational domain Ω is divided into $\Omega_1 = (0.0, 1.5) \times (1.0, 2.0)$ and $\Omega_2 = (0.0, 1.5) \times (0.0, 1.5)$. The porous medium Ω_2 has an enclosure of low permeability $(0.5, 1.0) \times (0.25, 0.75)$ with $\boldsymbol{K} = 10^{-5}\mathbf{I}$ surrounded by an area with $\boldsymbol{K} = \mathbf{I}$. We consider the following boundary conditions:

$$\boldsymbol{u}_1(x,y) = \begin{cases} (-(y-2.0)(y-1.0), 0.0), & \text{for } x = 0.0 \\ (-0.5(y-2.0)(y-1.0), 0.0), & \text{for } x = 1.5 \\ (0.0, 0.0) & \text{otherwise} \end{cases}$$

The Dirichlet and Neumann boundary conditions are chosen to be zero and the data functions $\boldsymbol{f}_1$ and f_2 are also zero. The expected solution is a parabolic flow from left to right in the free flow domain with some fluid percolating into the porous medium. We compare the computational time taken to solve the fully coupled problem and the two-grid method. First we solve the problem using the two-grid method, this requires solving the problem on a coarse mesh. The choice of a coarse mesh for this problem is $H = \frac{1}{8}$. The coarse mesh consists of 854 triangular elements. The choice of MINI elements in Ω_1 and the NIPG method of order two with zero penalty in Ω_2 results in a fully coupled problem of size 3995×3995 which takes 6.25 s to solve. Next we refine the coarse mesh twice to obtain a fine mesh with 13,664 triangular elements $(h = \frac{1}{32})$. On the fine mesh we solve a decoupled Navier–Stokes problem of size $26{,}935 \times 26{,}935$ and a Darcy problem of size $36{,}000 \times 36{,}000$. It takes 2.12×10^2 s to solve the resulting decoupled problems. In Fig. 4 we compare the solutions obtained from solving the fully coupled problem on the fine mesh. It is clear that both solutions capture the flow well and are comparable. The fully coupled problem for this case takes 1.904×10^3 s to solve. The two-grid method is 8 times faster than the fully coupled method in this example and gives an accurate solution.

6. Conclusion

We have presented a two-grid method for the coupled Navier–Stokes and Darcy problem. The method naturally yields decoupled and linear problems for the surface and subsurface regions. We have shown that the two-grid method is computationally efficient: the accuracy of the solution is comparable to fully coupled solution, and the CPU times are significantly smaller. Using the flexibility of discontinuous Galerkin methods in the subsurface, we obtained a more accurate solution by locally increasing the order of approximation.

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